

IB9190: Advanced Analytics — Group Assignment Report

FIFA World Cup 2026: Group Formation, Letter Assignment & Stadium Allocation

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1. Data Collection & Spectator Index

Following Ghoniem et al. (2017), four categories of data are required: (i) the 48 qualifying nations with FIFA points and confederation; (ii) four balanced pots; (iii) the 16-stadium match schedule; and (iv) a spectator index for each nation. All data are presented in full in the Appendices.

1.1 Nations, FIFA Points & Pots

42 confirmed teams were sourced from FIFA (2026c), with the 6 remaining vacancies filled by the highest-ranked team from each open qualification group (FIFA, 2026b; INSIDEFIFA, 2026). Canada, Mexico, and the USA are automatically seeded into Pot 1 as host nations. The remaining 45 teams are ranked by FIFA points and assigned to four pots of 12, with Pot 1 comprising the highest-ranked nations and Pot 4 the lowest (see Appendix Table 1).

1.2 Stadiums & Schedule

Stadium capacities and the official match schedule were collected from FIFA (2025; 2026a). The tournament uses 16 venues: 11 in the USA, 2 in Canada, and 3 in Mexico. The schedule maps each match to a specific stadium and date, enabling row-set construction for the letter assignment and stadium allocation models (see Appendix Table 2 for the full schedule). Since FIFA does not officially assign pot numbers (1–4) based on ranking points, the pot numbers allocated to each country are derived from our classification established in Section 1.1 (Nations, FIFA Points & Pots).

1.3 Spectator Index

The spectator index I measures the expected share of overseas supporters for each nation, scaled to 100. Host nations (USA, Mexico, Canada) are fixed at 100. For 24 nations participating in both Qatar 2022 and 2026, figures are taken directly from Ghoniem et al. (2017), excluding Qatar, Iran, and Egypt whose proximity to the 2022 host artificially inflated their values. For the remaining nations, the index is estimated as:

$$I_{target} = \min(100, I_{ref} * (G_{target} / G_{ref}))$$

where I_{ref} is the spectator index of a representative reference nation in the same confederation (CAF: Algeria; CONMEBOL: Brazil; UEFA: Germany; AFC: Japan; CONCACAF: Mexico; OFC: Australia), and G denotes 2026 GDP per capita from IMF (2025) projections. Scotland is equated to England (both UK); Curaçao uses World Bank (2026) GDP data. Full results are in Appendix Table 4.

2. Group Formation Model

The Group Formation Model assigns 48 qualified teams to 12 balanced groups by maximising the minimum group FIFA-points total — a Chebyshev objective that compresses disparity in group strength.

2.1 Model Formulation

Let N be the set of 48 nations, $G = \{1, \dots, 12\}$ the set of groups, and P_k the set of nations in Pot k ($k = 1, 2, 3, 4$). Let p_i denote FIFA points for nation i , and c_i its confederation. The binary decision variable $x_{ig} = 1$ if nation i is assigned to group g , and $w \geq 0$ is the Chebyshev variable representing the minimum group total.

Objective:

$$\text{Maximise } w$$

Subject to:

- Each team assigned exactly once: $\sum_{g \in G} x_{ig} = 1, \forall i \in N$
- Each group has exactly four teams: $\sum_{i \in N} x_{ig} = 4, \forall g \in G$
- One team per pot per group: $\sum_{i \in P_k} x_{ig} = 1, \forall g \in G, k = 1, \dots, 4$
- UEFA cap (at most 2 per group): $\sum_{i \in C^{UEFA}} x_{ig} \leq 2, \forall g \in G$
- Non-UEFA confederations (at most 1): $\sum_{i \in C^c} x_{ig} \leq 1, \forall g \in G, c \neq UEFA$
- Group points definition: $S_g = \sum_{i \in N} p_i * x_{ig} \forall g \in G$
- Chebyshev linking constraint: $w \leq S_g, \forall g \in G$

- Binary and non-negativity: $x_{ig} \in \{0,1\}, w \geq 0$

2.2 Modifications for 2026

The original model (Ghoniem et al., 2017) covered 32 teams across 8 groups with pots of 8. For 2026, the model is extended to 12 groups with 4 pots of 12. The UEFA allocation increases to 16 teams, making the cap of at most 2 UEFA teams per group the binding confederation constraint. Symmetry-breaking constraints fix each Pot 1 team (in descending FIFA points order) to a distinct group, reducing equivalent symmetric solutions without affecting feasibility or optimality.

2.3 Results & Comparison with FIFA Draw

The results demonstrate that the Chebyshev optimisation model substantially improves competitive balance compared to the FIFA-style formation. Under the baseline formation, the minimum group total was 6,141.06 FIFA points, with a wide dispersion reflected by a range of 425.85 and a standard deviation of 114.46. In contrast, the optimised model increased the minimum group total to 6,337.03, an improvement of 195.97 points (approximately 3.2%), thereby strengthening the weakest group. More importantly, the spread of group strengths was dramatically reduced: the range decreased to 90.87 and the standard deviation fell to 26.83. The maximum deviation from the average group strength was also reduced from 213.98 to 72.86. These results confirm that maximising the minimum group total effectively compresses disparities across groups, producing a significantly more balanced tournament structure while fully respecting pot and confederation constraints. Full group composition is given in Appendix Tables 4 and 5.

It is important to note a limitation of the Chebyshev objective: because the model only maximises the minimum group total w , the weakest group is guaranteed to be consistent across runs, but the composition of the remaining 11 groups is not uniquely determined. Multiple feasible solutions may share the same optimal value of w while distributing teams differently across the other groups, meaning that re-running the solver can yield different group formations. To address this and produce a fully stable, reproducible solution, an alternative or additional objective could be introduced — for example, minimising the range between the highest and lowest group totals (i.e. minimise $S_{\max} - S_{\min}$). This would constrain both ends of the distribution simultaneously, significantly reducing the solution space and ensuring a unique, balanced formation across all groups.

3. Group-Letter Assignment Model

Given the 12 balanced groups, the Group-Letter Assignment Model assigns each group to a letter (A–L) to maximise the spread of row-set popularity scores across the 16 stadium date-slots defined by the official schedule.

3.1 Why $w_{\max}$ is the Maximum Row-Set Popularity

In the Group-Letter Assignment Model, every stadium row-set r has an associated value called row-set popularity, denoted y_r . A row-set is a group of matches scheduled within the same stadium slot; its popularity depends on the spectator indices of the nations playing in those matches (following the calculation method in Ghoniem et al., 2017). Formally, y_r represents the total popularity of all matches in row-set r . The model introduces two bounding variables: $w_{\min}$, the lowest row-set popularity across all row-sets, and $w_{\max}$, the highest. Conceptually, $w_{\max} = \max_r(y_r)$ — that is, it is the popularity of the most attractive stadium day in the entire schedule.

To enforce this formally, the model introduces a binary selector variable $v_r \in \{0,1\}$ with the constraint $\sum_r v_r = 1$, which activates exactly one row-set. The constraint $w_{\max} \geq y_r \forall r$ ensures $w_{\max}$ cannot fall below any row-set popularity. The big-M constraint $w_{\max} \leq y_r + M(1 - v_r)$ then forces $w_{\max}$ to equal y_r precisely for the selected (activated) row-set. Here, we set $M=20$, given that the maximum popularity for a single match is 4, and a stadium can host at most 5 matches. Together, these three constraints guarantee that $w_{\max}$ takes the value of the highest y_r at optimality. Without the binary selection mechanism, the upper bound would be loose — $w_{\max}$ could be set arbitrarily high without being anchored to any actual row-set — making the objective unbounded. Hence the formulation correctly identifies and enforces the true maximum row-set popularity.

3.2 Model Formulation

Let $G = \{1, \dots, 12\}$ be groups, $L = \{A, \dots, L\}$ letters, and $R = \{1, \dots, 16\}$ row-sets. Let I_{rl} denote the spectator-index contribution of letter l to row-set r (derived from the match schedule). The binary variable $z_{gl} = 1$ if group g is assigned letter l ; $w_{\min}, w_{\max} \geq 0$ are the minimum and maximum row-set popularities; and $v_r \in \{0,1\}$ is the row-set selector.

Objective:

$$\text{Maximise } w_{\min} + w_{\max}$$

Subject to:

- One-to-one assignment: $\sum_l z_{g,l} = 1; \sum_g z_{g,l} = 1; \forall g \in G; \forall l \in L$
- Row-set popularity: $y_r = \sum_g I_{r,l} * z_{g,l}, \forall r \in R$
- Lower bound: $w_{\min} \leq y_r, \forall r \in R$
- Upper bound: $w_{\max} \geq y_r, \forall r \in R$
- Big-M selector (tightening): $w_{\max} \leq y_r + M(1 - v_r), \forall r \in R$
- Selector constraint: $\sum_r v_r = 1$
- Binary and non-negativity: $z_{g,l}, v_r \in \{0,1\}; w_{\min}, w_{\max} \geq 0$

3.3 Modifications for 2026

The expansion to 12 groups requires a 12×12 one-to-one assignment covering letters A–L. The row-set count increases to 16 under the 48-team schedule. The single-host pre-assignment constraint from Qatar 2022 is removed, as the 2026 tournament has three host nations whose groups cannot be pre-fixed without violating pot constraints.

3.4 Results & Analysis

The optimal solution yields $w_{\min} = 7.995$, $w_{\max} = 16.928$, and an objective value of 24.923. Compared to the temporary alphabetical baseline ($w_{\min} = 5.308$, $w_{\max} = 15.995$, range = 10.687), optimisation raises the minimum row-set popularity by 50.6% while maintaining a strong peak (16.928). The range narrows from 10.687 to 8.933, delivering a more commercially and competitively stable schedule across all 16 stadium date-slots.

4. Stadium Assignment Model

The final model assigns each of the 16 row-sets to one of the 16 stadiums to maximise total popularity value, defined as the product of stadium capacity and row-set popularity — directly building on the outputs of Questions 2 and 3.

4.1 Model Formulation

Let $R = \{1, \dots, 16\}$ be row-sets with popularity w_r , and $S = \{1, \dots, 16\}$ stadiums with seating capacity c_s . The binary decision variable $x_{rs} = 1$ if row-set r is assigned to stadium s , 0 otherwise.

Objective:

$$\text{Maximise } \sum_{r \in R} \sum_{s \in S} c_s * w_r * x_{r,s}$$

Subject to:

- Each stadium hosts exactly one row-set: $\sum_r x_{r,s} = 1, \forall s \in S$
- Each row-set is assigned to one stadium: $\sum_s x_{r,s} = 1, \forall r \in R$
- Binary restriction: $x_{r,s} \in \{0,1\}$

This is a 16×16 Linear Assignment Problem (LAP) with 256 binary variables and 32 constraints, solved to global optimality using GLPK via the Pyomo modelling interface.

4.2 Modifications for 2026

Ghoniem et al. (2017) solved an 8×8 LAP for Qatar's 8 venues. For 2026 the model scales to 16×16 reflecting the 16-stadium, 16-row-set structure. The objective coefficient — capacity × row-set popularity — is otherwise identical in structure.

4.3 Results & Analysis

The solver confirms a globally optimal total popularity value of 9,561,316 against a naive baseline (row-set r assigned to stadium $s = r$) of 9,441,244 — an improvement of 1.27%. The dominant assignment is Row-Set 7 to MetLife Stadium ($w = 16.928$, capacity 78,576), generating a pair value of 1,330,135 (13.9% of the total). Row-Set 9 maps to Estadio Azteca (782,453), reflecting strong Latin American demand. The two lowest-popularity row-sets (12 and 3) map to the two smallest venues (BMO Field and BC Place), avoiding wasted capacity. Full results are in Appendix Table 6.

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Appendices

Appendix Table 1: Nation Information — 48 Qualifying Nations with Confederation, FIFA Points, and Pot Assignment

Nation	Code	Confederation	FIFA Points	Pot
Spain	ESP	UEFA	1877.18	1
Argentina	ARG	CONMEBOL	1873.33	1
France	FRA	UEFA	1870.00	1
England	ENG	UEFA	1834.12	1
Brazil	BRA	CONMEBOL	1760.46	1
Portugal	POR	UEFA	1760.38	1
Netherlands	NED	UEFA	1756.27	1
Morocco	MAR	CAF	1736.57	1
Belgium	BEL	UEFA	1730.71	1
United States	USA	CONCACAF	1681.88	1
Mexico	MEX	CONCACAF	1675.75	1
Canada	CAN	CONCACAF	1559.15	1
Germany	GER	UEFA	1724.15	2
Croatia	CRO	UEFA	1716.88	2
Senegal	SEN	CAF	1706.83	2
Italy	ITA	UEFA	1702.06	2
Colombia	COL	CONMEBOL	1701.30	2
Uruguay	URU	CONMEBOL	1672.62	2
Switzerland	SUI	UEFA	1654.69	2
Japan	JPN	AFC	1650.12	2
Iran	IRN	AFC	1617.02	2
Denmark	DEN	UEFA	1616.75	2
South Korea	KOR	AFC	1599.45	2
Ecuador	ECU	CONMEBOL	1591.73	2
Austria	AUT	UEFA	1585.51	3
Turkey	TUR	UEFA	1582.69	3
Australia	AUS	AFC	1574.01	3
Algeria	ALG	CAF	1560.91	3
Egypt	EGY	CAF	1556.71	3
Ukraine	UKR	UEFA	1557.47	3
Norway	NOR	UEFA	1553.14	3
Panama	PAN	CONCACAF	1539.47	3
Ivory Coast	CIV	CAF	1522.48	3
Scotland	SCO	UEFA	1506.77	3
Paraguay	PAR	CONMEBOL	1501.50	3
Tunisia	TUN	CAF	1479.04	3
DR Congo	COD	CAF	1468.22	4
Uzbekistan	UZB	AFC	1462.03	4
Qatar	QAT	AFC	1454.96	4
Iraq	IRQ	AFC	1436.94	4
South Africa	RSA	CAF	1432.76	4
Saudi Arabia	KSA	AFC	1429.48	4

Nation	Code	Confederation	FIFA Points	Pot
Jordan	JOR	AFC	1388.93	4
Cape Verde	CPV	CAF	1370.49	4
Ghana	GHA	CAF	1351.09	4
Curaçao	CUW	CONCACAF	1302.70	4
Haiti	HAI	CONCACAF	1294.49	4
New Zealand	NZL	OFC	1279.25	4

Appendix Table 2: Stadium Schedule — Match Assignments by Stadium and Day

Row ID = Stadium ID. Cell entries show Group Letter + Pot numbers of competing teams (e.g. A12 = Group A, Pot 1 vs Pot 2). Pot numbers derived from Section 1.1 classification.

ID	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10	Day 11	Day 12	Day 13	Day 14	Day 15	Day 16	Day 17
1			D24					B13			G34			B12		G14	
2		B14					L34			E13			L23			I24	
3	A13						K23							A14			
4				F34						F23				A23			
5	A24							A12					K24			H12	
6			C12			I12						I23			E12		L13
7				F12			L12					J12			F24		J14
8						J13				E24					F13		J23
9				E14			K14			F14			K13			H34	
10					H14			A34			H13			C24			K34
11		D13			G24			B24			G12				D14		
12				E23					C14			I14			E34		L24
13					G13				D12					B34		G23	
14			B23			J24			D34			J34			D23		
15			C34			I34			C23				L14			I13	
16					H23						H24			C13			K12

Appendix Table 3: Spectator Index for All 48 Nations

Nation	Spectator Index (%)	Source / Note
Spain	51	2022 Qatar Data
Argentina	54	2022 Qatar Data
France	51	2022 Qatar Data
England	100	2022 Qatar Data
Brazil	94	2022 Qatar Data
Portugal	34	2022 Qatar Data
Netherlands	55	2022 Qatar Data
Morocco	41	Reference: Algeria
Belgium	38	2022 Qatar Data
Germany	63	2022 Qatar Data
Croatia	6	2022 Qatar Data
Senegal	17	Reference: Algeria
Italy	51	2022 Qatar Data
Colombia	77	Reference: Brazil
United States	100	Host nation
Mexico	100	Host nation
Uruguay	9	2022 Qatar Data
Switzerland	100	Reference: Germany (capped at 100)
Japan	29	2022 Qatar Data
Iran	3	Reference: Japan
Denmark	85	Reference: Germany
South Korea	13	2022 Qatar Data
Ecuador	66	Reference: Brazil
Austria	65	Reference: Germany
Turkey	18	Reference: Germany
Australia	77	2022 Qatar Data
Algeria	50	2022 Qatar Data
Canada	100	Host nation
Egypt	31	Reference: Algeria
Norway	100	Reference: Germany (capped at 100)
Panama	100	Reference: Mexico (capped at 100)
Ukraine	7	Reference: Germany
Ivory Coast	27	Reference: Algeria
Scotland	100	Reference: England (capped at 100)
Paraguay	4	2022 Qatar Data
Tunisia	50	2022 Qatar Data
DR Congo	7	Reference: Algeria
Uzbekistan	3	Reference: Japan
Qatar	61	Reference: Japan
Iraq	5	Reference: Japan
South Africa	58	Reference: Algeria
Saudi Arabia	25	Reference: Japan
Jordan	4	Reference: Japan
Cape Verde	51	Reference: Algeria
Ghana	23	Reference: Algeria

Nation	Spectator Index (%)	Source / Note
Curaçao	100	Reference: Mexico (capped at 100)
Haiti	14	Reference: Mexico
New Zealand	55	Reference: Australia

Appendix Table 4: Optimal Group Formation Results (12 Groups)

Grp	Pot 1	Pot 2	Pot 3	Pot 4	Total Points
1	United States	Colombia	Turkey	Uzbekistan	6,427.9
2	Spain	Switzerland	Egypt	Haiti	6,383.1
3	Argentina	Japan	Algeria	New Zealand	6,363.6
4	Belgium	Croatia	Tunisia	Saudi Arabia	6,356.1
5	England	Ecuador	Ukraine	Cape Verde	6,353.8
6	Portugal	Senegal	Australia	Curaçao	6,343.9
7	Netherlands	South Korea	Norway	South Africa	6,341.6
8	France	Iran	Paraguay	Ghana	6,339.6
9	Brazil	Denmark	Scotland	Qatar	6,338.9
10	Morocco	Uruguay	Panama	Jordan	6,337.6
11	Mexico	Italy	Ivory Coast	Iraq	6,337.2
12	Canada	Germany	Austria	DR Congo	6,337.0

Appendix Table 5: Group Popularity Values by Match Type

M1=Pot1 vs Pot2; M2=Pot1 vs Pot3; M3=Pot1 vs Pot4; M4=Pot2 vs Pot3; M5=Pot2 vs Pot4; M6=Pot3 vs Pot4

Grp	M1 (P1v2)	M2 (P1v3)	M3 (P1v4)	M4 (P2v3)	M5 (P2v4)	M6 (P3v4)	Total
1	3.65	2.77	2.55	1.89	1.60	0.42	12.86
2	3.27	1.64	1.30	2.96	2.71	0.90	12.78
3	1.66	2.08	2.18	1.58	1.68	2.10	11.28
4	0.88	1.76	1.26	1.12	0.62	1.50	7.13
5	3.49	2.61	3.27	1.46	2.34	1.17	14.32
6	1.01	2.22	3.01	1.87	2.75	3.66	14.52
7	1.36	3.11	2.26	2.27	1.42	3.16	13.57
8	1.09	1.10	1.48	0.15	0.53	0.54	4.88
9	3.59	3.91	3.11	3.78	2.94	3.42	20.75
10	1.00	2.83	0.91	2.19	0.26	2.09	9.29
11	3.27	2.91	2.57	1.56	1.11	0.63	12.04
12	3.45	3.48	2.60	2.57	1.40	1.44	14.94

Appendix Table 6: Optimal Stadium Assignment — Row-Sets to Stadiums (sorted by row-set popularity)

Row-Set	Stadium	w_r	Capacity	Pair Value
7	MetLife Stadium	16.928	78,576	1,330,135
9	Estadio Azteca	10.753	72,766	782,453
8	AT&T Stadium	9.373	70,122	657,254
16	SoFi Stadium	9.358	69,650	651,785
10	Levi's Stadium	9.079	69,391	630,001
4	NRG Stadium	9.074	68,311	619,854
11	Arrowhead Stadium	8.810	67,513	594,790
2	Mercedes-Benz Stadium	8.566	67,382	577,194
13	Lincoln Financial Field	8.509	65,827	560,122
15	Lumen Field	8.465	65,123	551,266
6	Hard Rock Stadium	8.447	64,091	541,377
14	Gillette Stadium	8.387	63,815	535,216
1	Estadio BBVA	8.283	50,113	415,086
3	BC Place	8.194	48,821	400,039
5	Estadio Akron	8.131	44,330	360,447
12	BMO Field	7.995	44,315	354,298
TOTAL	9,561,316			

Optimal total popularity value: 9,561,316 | Naive baseline ($r = s$): 9,441,244 | Improvement: +1.27%